The retention time plot

Observed retention time against the retention time written into the report as predicted, for every row of the peptide report.

AT A GLANCE

Where	Peptide Level → RT Plot
Needs	A peptide report, loaded as the Peptide level in Data → Upload. The peptide and precursor levels are read from the long-format report on disk rather than from a wide matrix; every other page of this part needs an abundance column in that report, and this one does not. It reads two columns, Predicted.RT and RT, and needs no annotation either.
One point	One row of the report. Nothing is grouped or de-duplicated first, so a peptide identified in several runs, or at several charges, contributes one point for each.
Two renderers	**Type** Normal draws a static figure in one of three modes. Interactive draws a scatter that can be hovered and zoomed, and opens in its own tab.
Table	None.

1 What the plot answers

Both numbers on this figure come out of the report: RT is when a precursor was actually seen, Predicted.RT is when whatever produced the report expected to see it. The tool computes neither. It draws one against the other, so the question the page answers is how far the chromatography of this experiment departed from the run the prediction was made for.

A well-aligned experiment puts the whole cloud on the diagonal, narrow enough that its width is the scatter of single identifications rather than a property of the gradient. Departures come in two shapes, and they mean different things: a *bend* is systematic and affects every precursor in a retention window, while *individual points far off the line* are single identifications whose prediction and observation disagree.

> **A SMALL EXAMPLE**
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> If both axes are in minutes, a precursor predicted at 20 and observed at 23 lies above the identity line: it eluted three minutes later than predicted. A whole band with that displacement suggests a systematic mismatch worth checking, not that every identification is wrong. An indexed prediction and an observed time cannot be compared this way without calibration.

Reading it

- One point, one row of the report. In hexbin and density mode the points are binned and it is the colour of a cell, not a point, that is drawn — the colour bar gives how many rows fell into it.
- The identity line, $y = x$. It is drawn only with **Add Line** on. A point above it eluted later than predicted, a point below it earlier.
- A point far from the line: one identification whose observed and predicted times disagree. In a report of a few hundred thousand rows a scatter of these is expected.
- A systematic curve or step in the band. Every identifications in that retention window may be affected together. Check calibration, the prediction model, gradient and column conditions before interpreting individual departures.

> **EASY TO GET WRONG**
>
> The axes do not carry a unit, because the tool takes both columns as the report wrote them and converts neither. If the two columns were produced on different scales — one in minutes, one in seconds, or one indexed rather than timed — the cloud will lie along a straight line that is not the diagonal, and the identity line will look wrong when nothing about the chromatography is. Read the slope of the band before reading its width.

Confirm that both columns use the same units and calibration before treating the identity line as agreement. The application does not establish that from the column names alone.

2 Controls

Controls on this page

Control	What it does	Default	Where
Mode	How the points are drawn: Scatter Plot draws them individually, Hexbin Plot bins them into hexagons coloured by count, Density Plot bins them into a square grid coloured by count on a logarithmic scale. Both binned modes add a colour bar. Ignored when **Type** is Interactive.	Hexbin Plot	page
Type	Normal renders a static image. Interactive renders a scatter with hover text, zoom and pan, shown in a frame with its own **Open Interactive Plot** link instead of a download button.	Normal	page
Bins	The resolution of the binned modes: the hexagon grid size in Hexbin Plot, the number of bins per axis in Density Plot. It is held between 10 and 2000, and has no effect in Scatter Plot or in the interactive renderer.	1000	page
Random Seed	Fixes the sub-sample drawn when the report is larger than the renderer draws in full, so that the same figure is reproduced exactly. It has no effect below that size, and none in the binned modes.	69	page
Add Line	Draws the identity line $y = x$ across the data.	off	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm and DPI between 72 and 600, and a very large figure has its resolution reduced so that the image stays under 24 megapixels. Shown only while Normal is selected. The interactive plot has none: it fills its frame, Plotly's camera button saves a picture of it, and **Open Interactive Plot** opens it in a window of its own.	20 cm × 15 cm, 100	page
Download format	PNG, JPG, WEBP, SVG or PDF. Offered for the static renderer only.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Reference line	The identity line: shown or hidden, and its colour, dash pattern and width. Hiding it here removes the line even with **Add Line** on; **Add Line** decides whether there is a line at all, this decides what it looks like.	shown, red, dashed, width 1.5	appearance
Points	Colour, shape, size, opacity and outline of the dots. The plot asks for a diameter of 1 pt at opacity 0.2 in the static scatter and 4 pt at 0.35 in the interactive one, and the panel overrides both. The outline is off unless it is asked for.	the plot's own	appearance
Title, X axis, Y axis	Text, size, weight, colour and position of each label, and the size, rotation and colour of the tick labels. The plot writes Retention Time Plot, Predicted RT and Actual RT.	the plot's own	appearance
Font	One face for the whole figure, or one per text.	plot default	appearance
Frame and grid	Which of the four spines are drawn, and the grid behind the data.	all four spines	appearance
Background	Paper colour, transparency and the preset themes.	default	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here. The panel keeps one set of settings per plot, and this page has two renderers, so its defaults follow whichever of them **Type** has selected.

3 How it is computed

HOW IT IS COMPUTED

The columns Predicted.RT and RT are required by those exact names; a report without either draws nothing and says which is missing. Both are read as numbers, anything that cannot be read as one becomes NA, and a row is kept only if *both* of its values

survive. The surviving rows are the points: x is Predicted.RT, y is RT.
No grouping happens at any stage. One row of the report is one point, so a peptide seen in twelve runs at two charges contributes twenty-four points, and all runs are pooled into a single cloud. Nothing on this page splits by sample or by condition.

THE THREE STATIC MODES

Scatter Plot draws every point. Above 100 000 rows it first draws a random sample of 100 000 of them, chosen with **Random Seed**.
Hexbin Plot tiles the plane with hexagons whose grid size is **Bins**, clamped to $[10, 2000]$, and colours each by how many rows fell into it on a linear scale.
Density Plot bins the plane into a square grid of **Bins** cells per axis, again clamped to $[10, 2000]$, and colours each cell by its count on a logarithmic scale, so that a sparse tail is still visible beside a dense diagonal.
Both binned modes draw a colour bar; the scatter does not, because each of its marks is one row.

THE IDENTITY LINE

With **Add Line** on, a straight line is drawn from $(x_{\min}, x_{\min})$ to $(x_{\max}, x_{\max})$, where $x_{\min}$ and $x_{\max}$ are the smallest and largest Predicted.RT among the points drawn. It is the locus of perfect agreement: a precursor on it was observed exactly when it was predicted. The interactive renderer spans the same line over the smaller of the two minima and the larger of the two maxima, so its line reaches the corners of the frame.

THE INTERACTIVE RENDERER

Interactive always draws a scatter, whatever **Mode** says. Above 250 000 rows it draws a random sample of 250 000, again fixed by **Random Seed**. Each point carries hover text with the feature, the predicted and actual times to four decimals, their difference as Actual − Predicted, and the sample.
The feature is the first of Stripped.Sequence, Modified.Sequence, Precursor.Id, PrecursorID, Precursor or ProteinNames that the report has, failing that any column whose name contains *sequence*, and failing that a running number. The sample is the filename column reduced to the file's own name without its folders, truncated to 14 characters.

Mode and **Bins** apply to the static renderer only; with **Type** set to Interactive they are greyed out, because the interactive figure is always a scatter.

Edge cases

- Replicates are not a concept on this page: every run in the report is pooled into one cloud, and a single-run report is drawn exactly as a twelve-run one is.
- A row whose Predicted.RT or RT is missing, blank or non-numeric is dropped. It is dropped silently, and nothing on the figure says how many were.
- If no row has both values the page draws nothing and reports that no numeric values are available.
- If every point has the same predicted time, the identity line collapses to a single point and the axis to a single value; the figure is drawn, but there is nothing on it to read.
- The interactive renderer needs the plotly package. Where it is not installed the page says so and names the package.

4 Exporting

The static figure downloads at the width, height and resolution set on the page, which is why those three controls do not change what is on screen: the displayed plot is fitted to the window, the downloaded one to the numbers. Choose PDF for a figure that will be scaled or edited, PNG for a slide. A scatter of several hundred thousand points is large in PDF and slow to open; the binned modes are the better choice for a vector figure.

The interactive renderer has no download. It opens in its own tab, where its own toolbar saves the current view as a PNG. Its settings are also written into the summary report, which renders this page as a static figure whatever **Type** is set to here.

Behind this page

Function	What it produces
rt_vs_pred_rt_plot	Actual against predicted retention time, in scatter, hexbin or density mode. The package fixes the density mode at 200 bins per axis where the page passes **Bins**.
peptide_rt_plot	The static figure as the page renders it, including the appearance-panel settings. Its interactive twin is peptide_rt_interactive_html, which produces the scatter and its hover text.